Answers to quiz

Question 1

Q1(a)

The 37th percentile in the Unif(11, 36) distribution:

```
qunif(0.37, 11, 36)
```

```
[1] 20.25
```

Q1(b)

The 12th and 88th percentiles in the standard normal distribution:

```
qnorm(c(0.12, 0.88))      # One can also use  qnorm(c(0.12,0.88), 0, 1)
```

```
[1] -1.174987  1.174987
```

Not surprisingly, they have the same magnitude, but are of opposite sign.

Q1(c)

Though it is possible to wrap everything into a single, compound command, here I've done it in several steps:

```
mySample = rexp(600, rate=0.025)
mean(~mySample)                 # This assumes mosaic package is loaded
```

```
[1] 42.3416
```

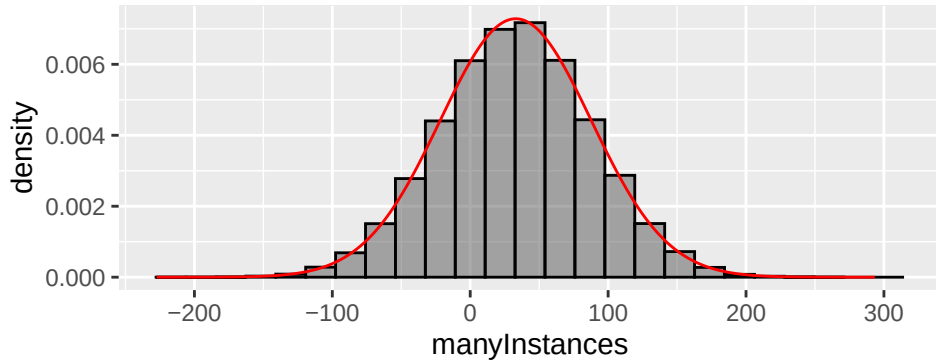
Answers, here, will vary, but should be fairly close to mine.

Question 2

Q2(a)

The "core" command provided produces one instance of a difference in scores. To see a distribution, we need to repeat this one command many times. Doing so reveals what values occur, and how often:

```r
manyInstances = replicate(100000, sum(rnorm(3,133,18)) - sum(rnorm(3,122,26)))
gf_dhistogram(~manyInstances, color="black", bins=25) |>
  gf_dist("norm", mean=33, sd=54.772, color="red")
```



Q2(b)

Define the random variable $D = B_1 + B_2 + B_3 - A_1 - A_2 - A_3$. According to theory, the expected value

$$\mathrm{E}(D) = \mathrm{E}(B_1) + \mathrm{E}(B_2) + \mathrm{E}(B_3) - \mathrm{E}(A_1) - \mathrm{E}(A_2) - \mathrm{E}(A_3) = 3(133) - 3(122) = 33.$$

Since scores are independent, variances are additive:

$$\mathrm{Var}(D) = \mathrm{Var}(B_1) + \mathrm{Var}(B_2) + \mathrm{Var}(B_3) + \mathrm{Var}(A_1) + \mathrm{Var}(A_2) + \mathrm{Var}(A_3) = 3(18)^2 + 3(26)^2 = 3000.$$

The standard deviation of the distribution, then, is $\sqrt{3000} \doteq 54.772$.

Our simulation, each time it is re-run, would produce a different (sample) mean and standard deviation, but they should be close to the theoretical values:

```r
mean(~manyInstances)
```

```
[1] 32.83124
```

```r
sd(~manyInstances)
```

```
[1] 54.81376
```

And, yes, they are.

Q2(c)

The distribution, produced in part (a), does look symmetric and bell-shaped. Since the values come from iid samples, drawn from normal populations, this is entirely expected, as sums/differences of independent normal r.v.s are, once again, normal.

Q2(d)

Angie wins every time $D = B_1 + B_2 + B_3 - A_1 - A_2 - A_3$ is negative. In our simulation, the proportion of times this happened is

```
prop(~(manyInstances < 0))
```

```
prop_TRUE
  0.27609
```

Our theory says $D \sim \text{Norm}(33, 54.772)$. (That is the red normal curve you see overlain on the density histogram in part Q2(a).) Employing the `pnorm()` command, the cumulative distribution function, we obtain a similar answer:

```
pnorm(0, 33, 54.772)
```

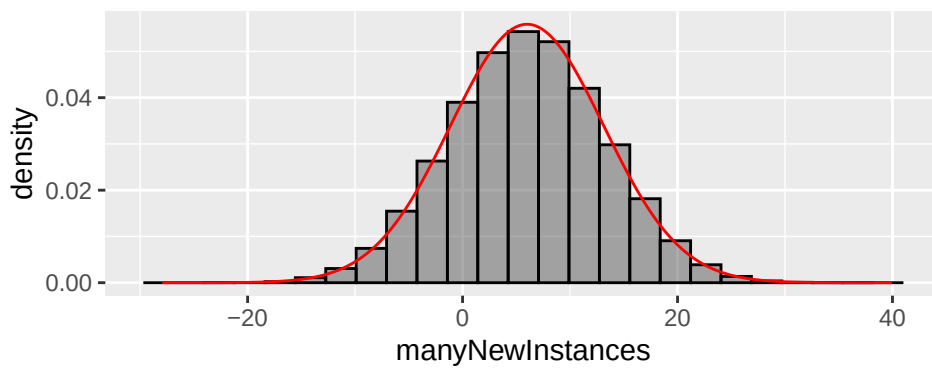
```
[1] 0.2734215
```

Question 3

Q3(a)

Altering the answer to Q2(a) as needed, we have

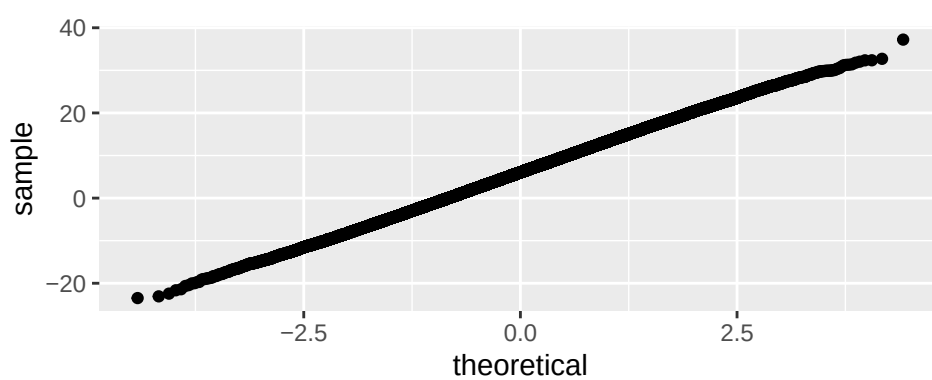
```
manyNewInstances = replicate(100000, mean(~runif(5,115,151)) - mean(~runif(5,106,148)))
gf_dhistogram(~manyNewInstances, color="black", bins=25) |>
  gf_dist("norm", mean=6, sd=sqrt(51), color="red")
```



Q3(b)

The histogram from part Q3(a) already appears symmetric and bell-shaped. It shouldn't surprise us if the normal quantile plot produces a near-straight line.

```
gf_qq(~manyNewInstances)
```



That looks pretty straight, verifying the values of $\bar{X}_B - \bar{X}_A$ are approximately normal.

Q3(c)

There is the empirical answer of how often Angie wins, which we ascertain from our simulated data:

```
prop(~(manyNewInstances < 0))
```

```
prop_TRUE
  0.20295
```

We have formulas for the mean and variance in a Unif(a, b) population:

$$\mu = \frac{1}{2}(a+b) \qquad \text{and} \qquad \sigma^2 = \frac{(b-a)^2}{12}.$$

So, each of Angie's games have

$$\mathrm{E}(A_i) = \frac{106+148}{2} = 127 \qquad \text{and} \qquad \mathrm{Var}(A_i) = \frac{(148-106)^2}{12} = 147,$$

and each of Brad's games have

$$\mathrm{E}(B_i) = \frac{115+151}{2} = 133 \qquad \text{and} \qquad \mathrm{Var}(B_i) = \frac{(151-115)^2}{12} = 108.$$

Now that game scores are being drawn from non-normal distributions, all the Central Limit Theorem guarantees is that,

$$\bar{X}_1 = \frac{1}{5}(B_1+B_2+B_3+B_4+B_5) \qquad \text{and} \qquad \bar{X}_2 = \frac{1}{5}(A_1+A_2+A_3+A_4+A_5)$$

will have distributions which are *more normal* than if the sample sizes were smaller than 5. But our evidence from part Q3(a) bolsters confidence that the distribution of differences is nearly normal. So, if we proceed as if $n = 5$ is large enough for normality for $\bar{X}_1$ and $\bar{X}_2$, individually, then

$$\bar{X}_1 \sim \mathrm{Norm}\left(133, \sqrt{\frac{108}{5}}\right), \qquad \bar{X}_2 \sim \mathrm{Norm}\left(127, \sqrt{\frac{147}{5}}\right),$$

and their difference

$$\bar{X}_1 - \bar{X}_2 \sim \mathrm{Norm}\left(6, \sqrt{\frac{108}{5} + \frac{147}{5}}\right) = \mathrm{Norm}(6, \sqrt{51}).$$

(That distribution was overlain on the density histogram in part Q3(a), and appears to fit the simulated differences quite well.) Let's use `pnorm()` and see how close this normal approximation's answer is to the empirical one:

```
pnorm(0, 6, sqrt(51))
```

```
[1] 0.2004071
```

If you compare with the above, the two answers are quite close, suggesting that our samples from uniform populations are of sufficiently large enough size that $(\bar{X}_1 - \bar{X}_2)$ has a near-normal distribution.